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KING'S College LONDON

Non-monotonic Reasoning, Argumentation and Dialogue

Week 3 Learning Outcomes

By the end of this week, you will be able to:

- LO1: Identify how to use argument game proof theories to establish whether an argument is in the grounded labelling of an argument framework

- LO2: Identify how to use argument game proof theories to establish whether an argument is in a preferred labelling of an argument framework

- LO3: Describe how argument game proof theories can be generalized to dialogues for distributed non-monotonic reasoning

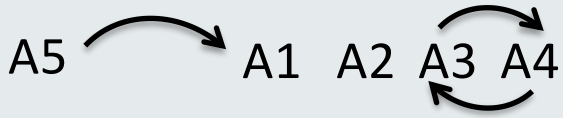
- LO4: Describe the importance of these dialogues given recent advances in Artificial Intelligence

Non-monotonic Reasoning as *Argumentation*

- We have seen how arguments can be constructed from a KB and related to each other by a binary attack relation whose success may depend on preferences amongst arguments

- The arguments constructed from a KB can then be organised into a directed graph in which the arguments are nodes and the successful attacks are directed binary relations amongst nodes

- We thus obtain a Dung Argumentation Framework (AF) < *Args,Attacks* >



Labelling Argumentation Frameworks

- Given an AF < *Args,Attacks* > of arguments related by attacks:

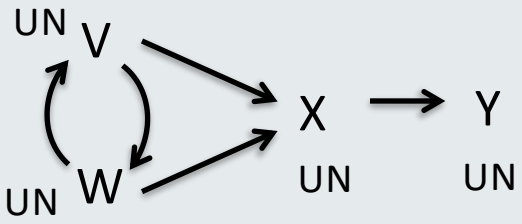
A labelling $\mathscr{L}$ results from applying a *total function LAB*(AF) *that assigns to each X* $\in$ *Args, either IN (winning), OUT (losing) or UN (undecided), and such that:*

1. *X is IN iff* $\forall$ *(Y,X)* $\in$ *Attacks* : *Y is OUT*
2. *X is OUT iff* $\exists$ *(Y,X)* $\in$ *Attacks* : *Y is IN*
3. *X is UN iff* $\neg\exists$ *(Y,X)* $\in$ *Attacks* such that *Y is IN* **and** $\exists$ *(Y,X)* $\in$ *Attacks* such that *Y is UN*

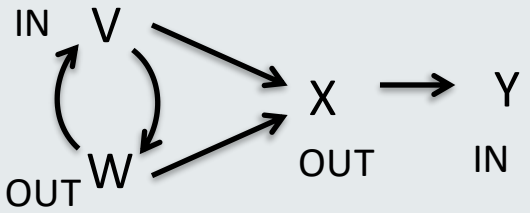
Labelling Argumentation Frameworks

- *LAB* may return many labellings (*LAB* is a one-to-many function)

$\mathscr{L}_1$ =



$\mathscr{L}_2$ =



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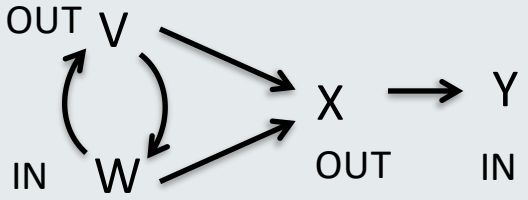

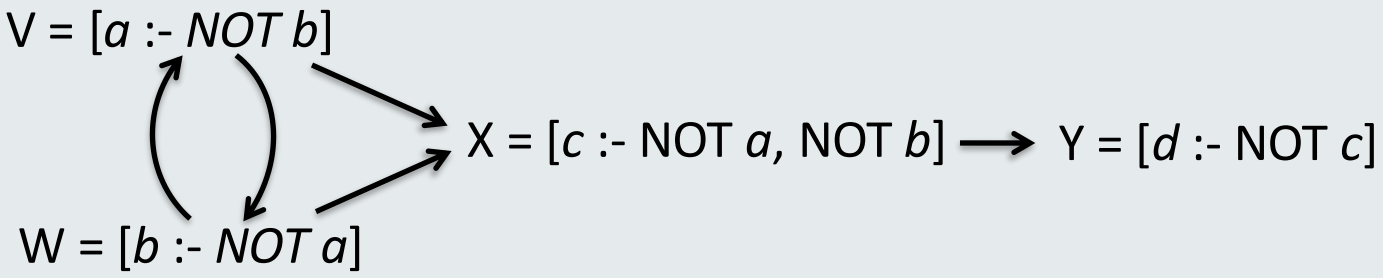

- A "preferred" labelling $\mathscr{L}$ is such that for all other labellings $\mathscr{L}^*$ the arguments made IN by $\mathscr{L}^*$ are not a superset of the arguments made IN by $\mathscr{L}$ (e.g., $\mathscr{L}_2$ and $\mathscr{L}_3$)

Labelling Argumentation Frameworks

- *LAB* may return many labellings (*LAB* is a one-to-many function)

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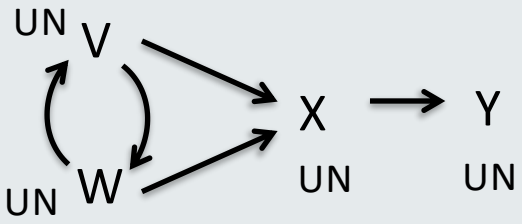

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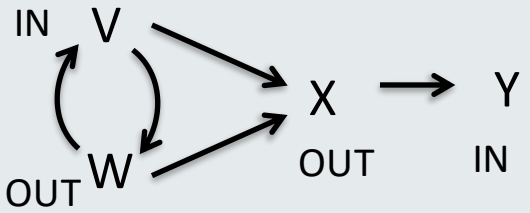

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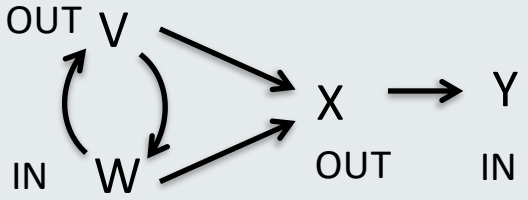

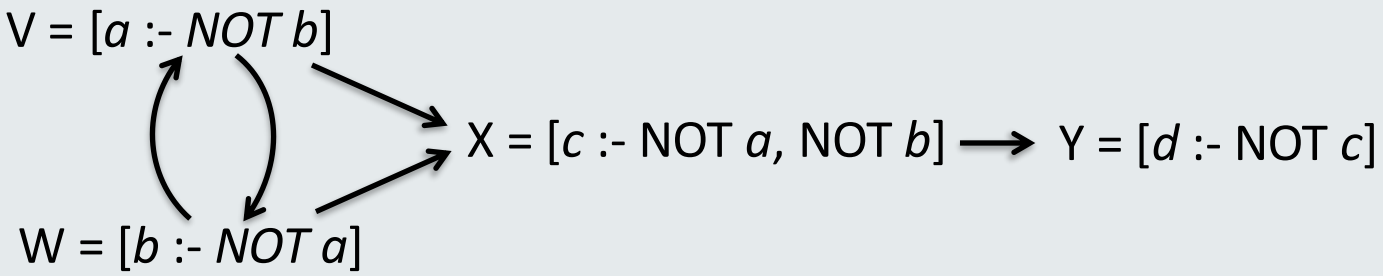

- **<u>The</u>** "grounded" labelling $\mathscr{L}$ is such that for all other labellings $\mathscr{L}^*$ the arguments made IN by $\mathscr{L}^*$ are not a subset of the arguments made IN by $\mathscr{L}$ (e.g., $\mathscr{L}_1$)

Week 3 Learning Outcomes

By the end of this week, you will be able to:

- LO1: Use argument game proof theories to establish whether an argument is in the grounded labelling of an argument framework

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Argument games – the basic idea

- Suppose you want to know whether X is IN (winning) in a preferred or grounded labelling of an AF.
- Two player game in which the *proponent* (PRO) *moves* X, and *opponent* (OPP) then moves an argument Y that attacks X (remember that these games are played by reference to an ***existing*** AF of arguments and attacks)
- PRO can then move a counter-attack Z to Y (ie., a Z that defends X)
- OPP then counter-attacks Z, and so on ....

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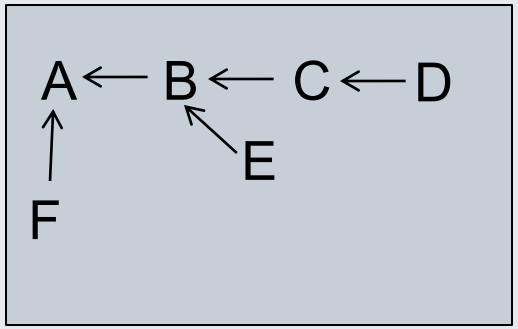
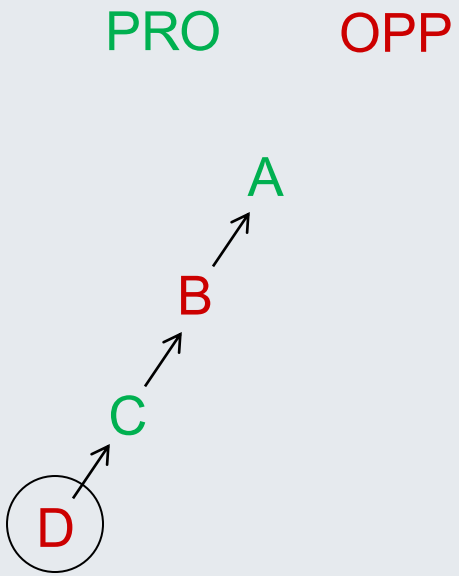
- PRO can then move a counter-attack Z to Y (ie., a Z that defends X)
- OPP then counter-attacks Z, and so on ....
- If at any stage a player is stuck and cannot move a counter-attack, she can backtrack to a previous move of her adversary, and try an alternative counter-attack
- If PRO *successfully* counter-attacks (in the sense that PRO is not in turn counter-attacked by OPP) **every move by OPP**, then PRO wins, else PRO loses.

Argument games – the basic idea

We want to know whether A is IN

AF =





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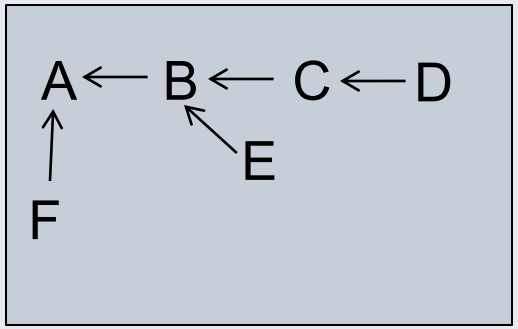

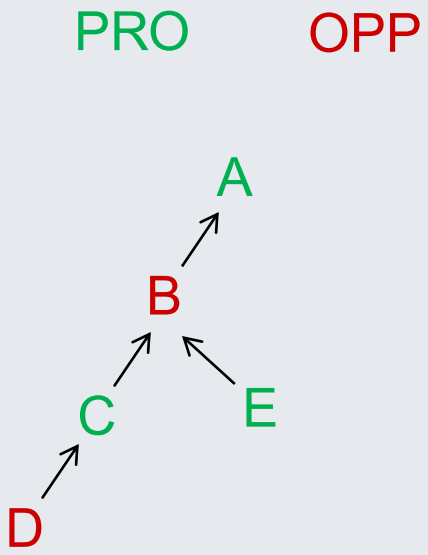

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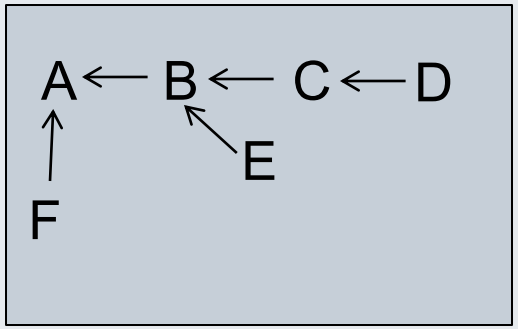

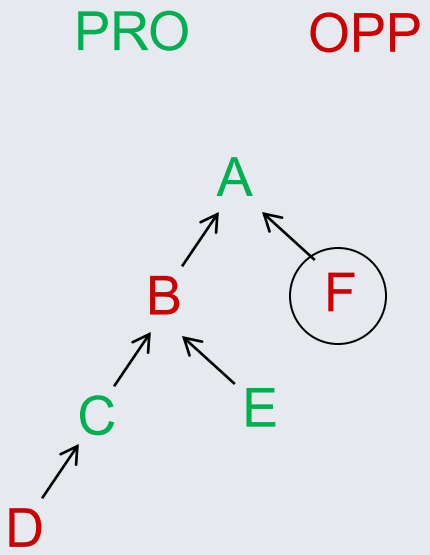

PRO loses because not every OPP move is counter attacked by a PRO move

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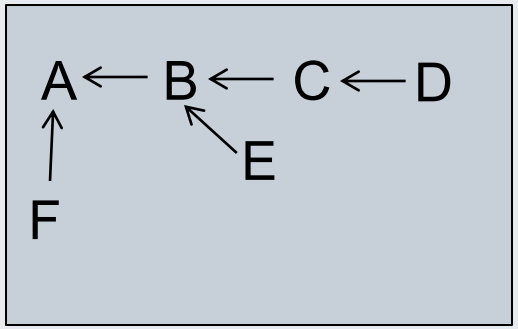

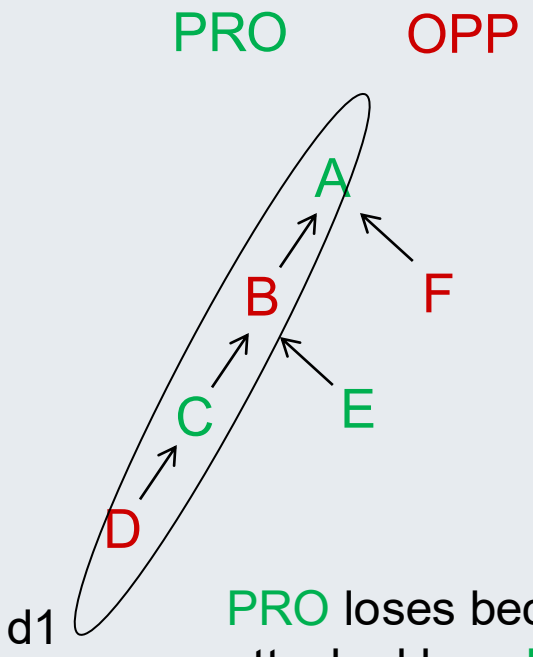

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Note that we have a **game tree** with three ***disputes***

Argument games – the basic idea

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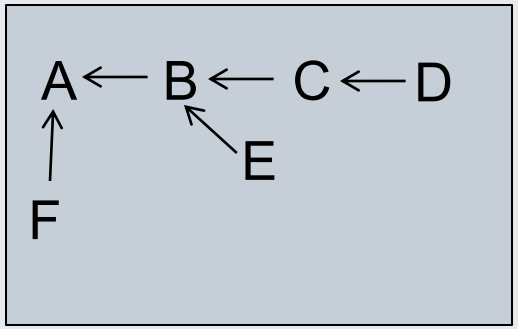

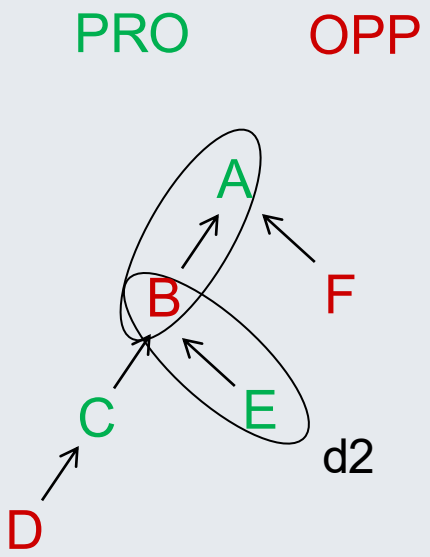

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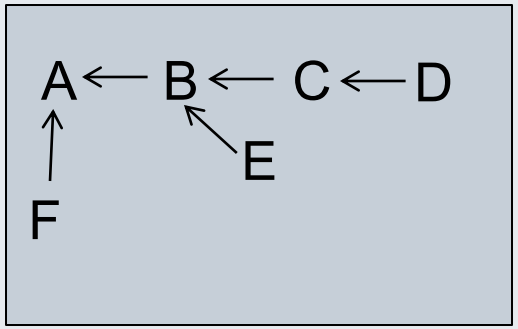
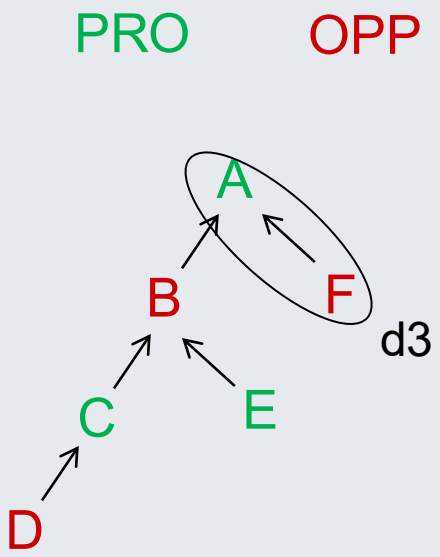




PRO loses because not every OPP move is counter attacked by a PRO move

Note that we have a **game tree** with three ***disputes***

Argument game rules

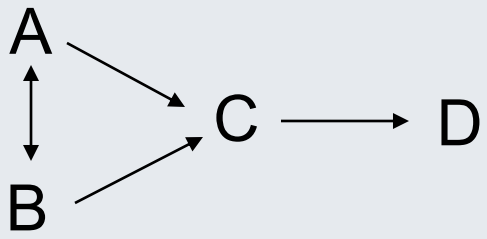
- But this is not the whole story. Depending on whether we are interested in whether an argument is IN in the **grounded** or a **preferred** labelling, the rules on what moves can be made (what moves are *legal*) vary

- In the *grounded game* (PRO tries to show that X is IN in the grounded labelling) ***PRO cannot repeat an argument in the same dispute***

- In the *preferred game* (PRO tries to show that X is in IN in a preferred labelling) ***OPP cannot repeat an argument in the same dispute***

Explaining the rules of the argument games

- Recall the AF :

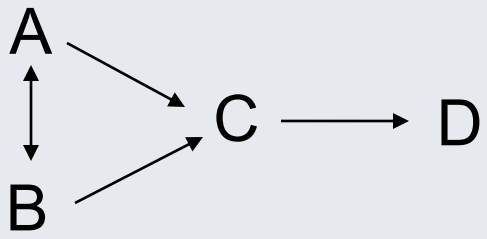


L1 = {A,D} and L2 = {B,D} are two preferred IN labellings

There are no IN arguments in the grounded labelling (∅ is the set of IN arguments in the grounded labelling)

Explaining the rules of the preferred game

- Recall the AF :



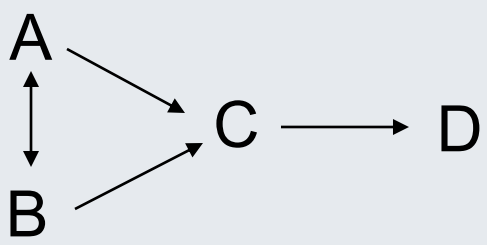
B A

L1 = {A,D} and L2 = {B,D} are two preferred IN labellings

- In each preferred labelling there is an argument ***that defends itself*** against an attack – A in L1 and B in L2

Explaining the rules of the preferred game

- Recall the AF :



E1 = {A,D} and E2 = {B,D} are preferred labellings

- Why the rule preventing OPP repeating in a preferred game ?

- Because : 1) otherwise the game may go on forever ; 2) PRO has already defended itself against OPP's repeated argument

Explaining the rules of the preferred game

AF =

$A \leftrightarrow B$

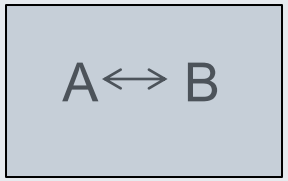
Is A in a preferred labelling ?

PRO OPP

A
↑
B
↑
A
↑
B
↑

Explaining the rules of the preferred game

AF =



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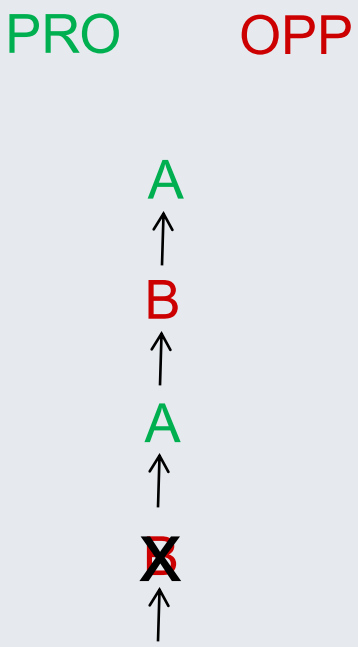
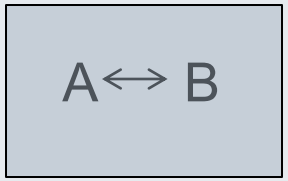


Explaining the rules of the preferred game

AF =



Is A in a preferred labelling ?

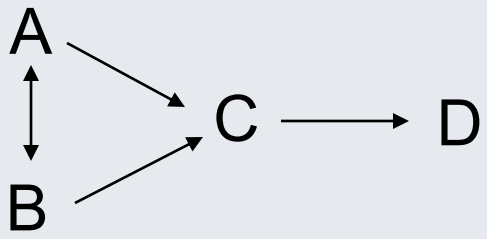
PRO OPP

A
↑
B
↑
A

- PRO wins – there is a preferred labelling {A} that contains A

Explaining the rules of the grounded game

- Recall the AF :



There are no IN arguments in the grounded labelling (∅ is the set of IN arguments in the grounded labelling)

- The grounded semantics places a higher burden of proof – one cannot in defending an argument X, either directly ***or indirectly*** self-defend with X. In an argument game this amounts to PRO not repeating X in a dispute.

Explaining the rules of the grounded game

AF =

$A \leftrightarrow B$

Is A in the grounded labelling ?

PRO OPP

A

↑

B

↑

A

Explaining the rules of the grounded game

AF =

$A \leftrightarrow B$

Is A in the grounded labelling ?

PRO OPP

A

↑

B

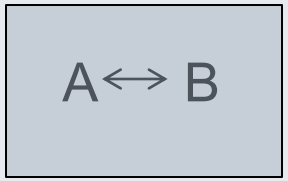
↑

~~A~~

X

Explaining the rules of the grounded game

AF =



PRO OPP

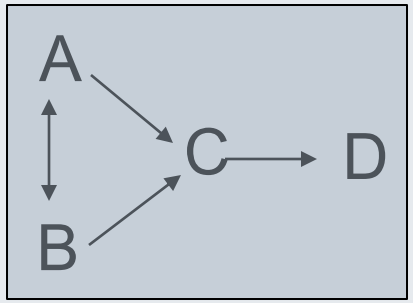
A
↑
B

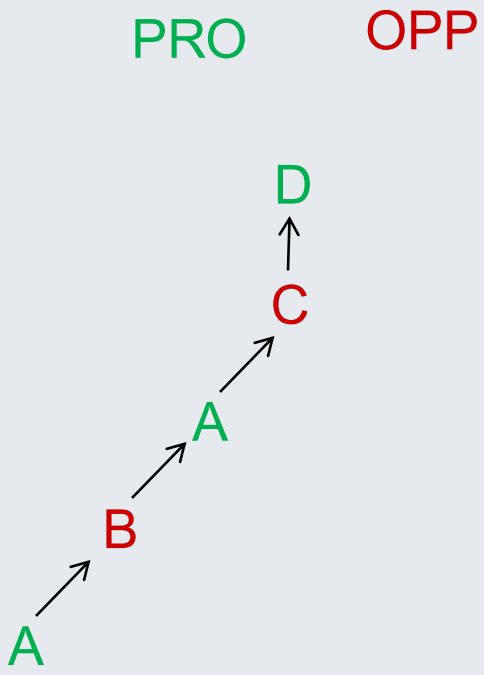
Is A in the grounded labelling ?

- PRO loses – the grounded labelling is {A = UN, B = UN}

Example – is D in a preferred labelling ?

AF =





PRO wins

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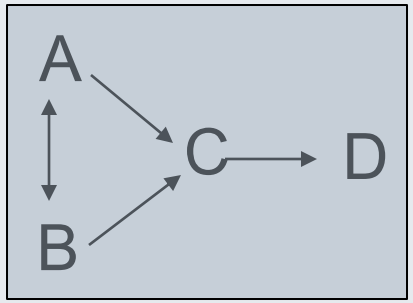
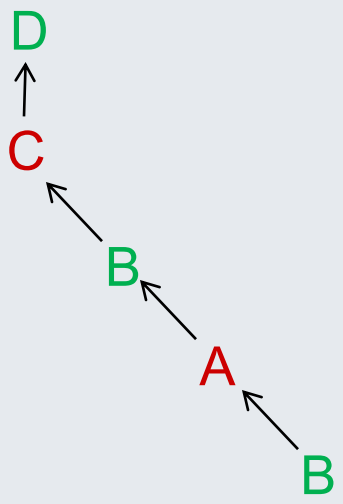


PRO OPP



PRO wins

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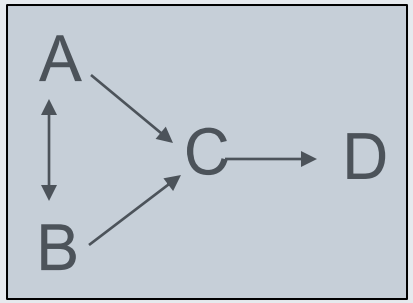

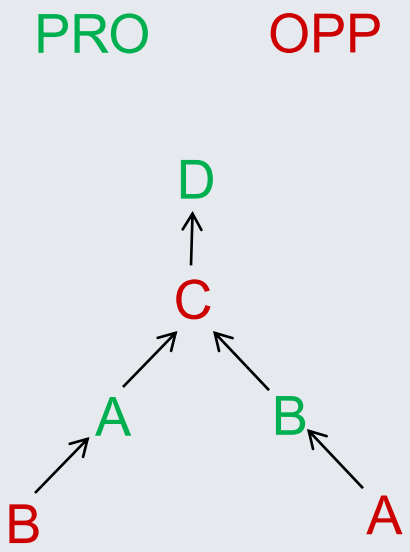

PRO loses

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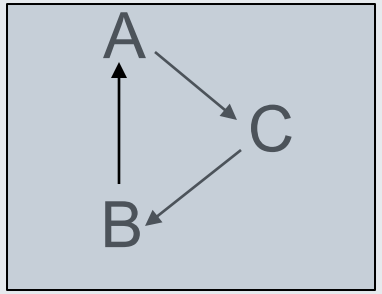

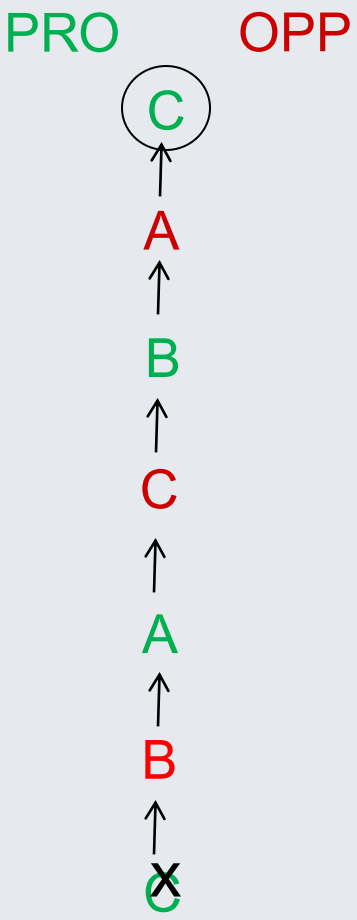

In the grounded game PRO cannot repeat an argument in the same dispute

Example – is D in the grounded labelling ?

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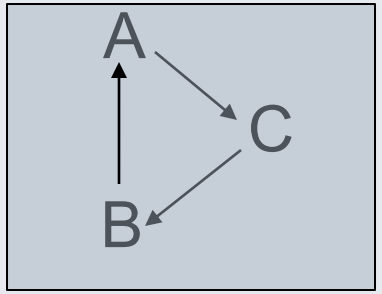


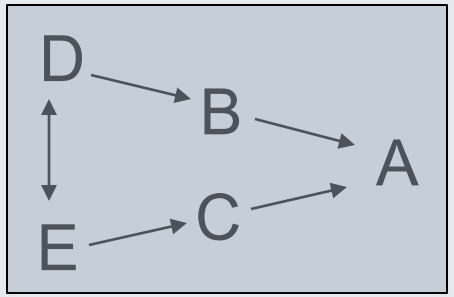
PRO OPP

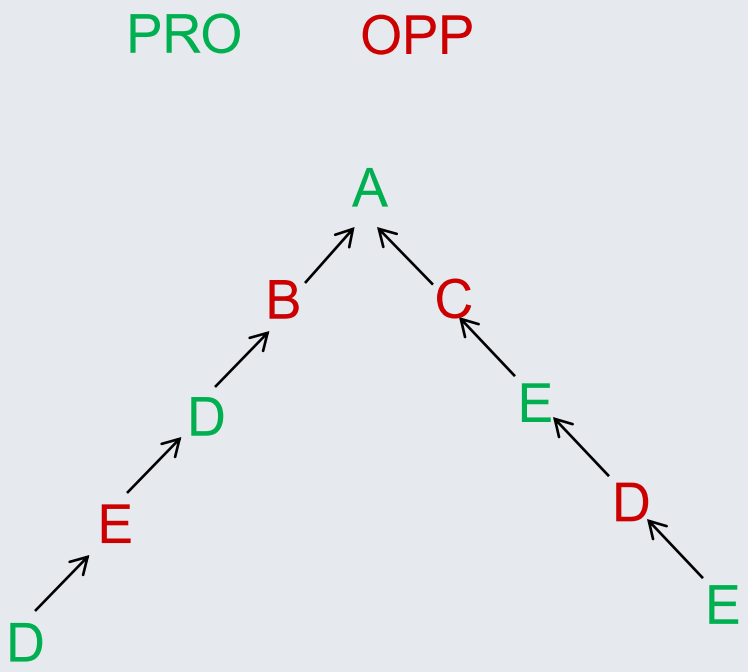
C
↑
A
↑
B
↑
C
↑
A
↑
B

PRO loses

Another Example – Is A in a preferred labelling ?

AF =





Should PRO win ?

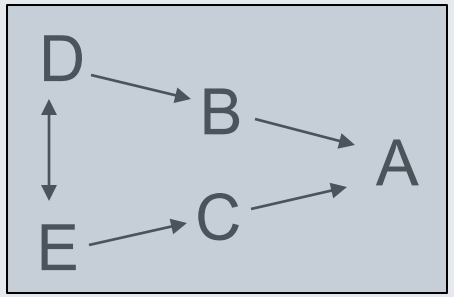
Argument games : moves by PRO

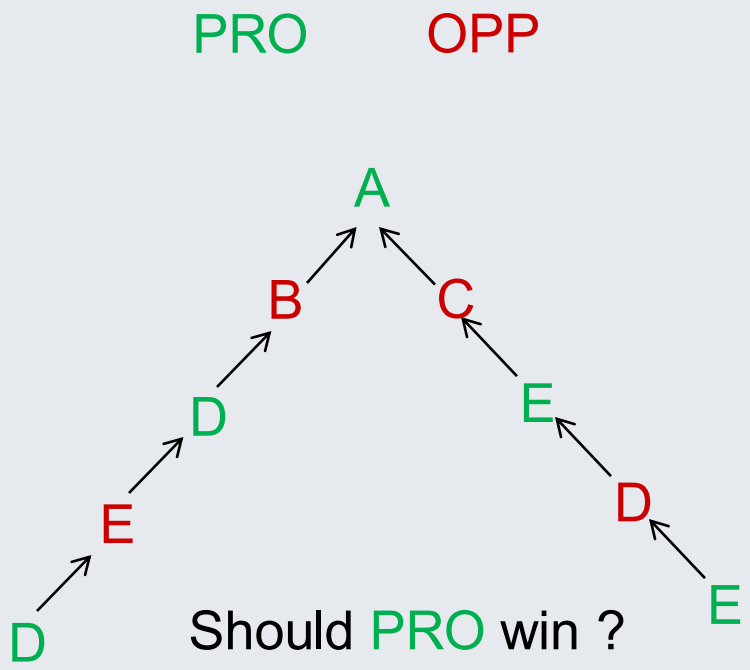
- In the grounded game, if PRO wins, then the arguments moved by PRO are a subset of the grounded labelling (are all IN in the grounded labelling)

- In the preferred game, if PRO wins, then the arguments moved by PRO are a subset of a preferred labelling (are all IN in a preferred labelling)

Another Example – Is A in a preferred labelling ?

AF =





Should PRO win ?

No, because D and E are used by PRO to defend A - but D and E attack each other !

Defining the Grounded Game

- X is in the grounded labelling of < *Args,Attacks* > iff

  There exists a game tree T with root X such that:

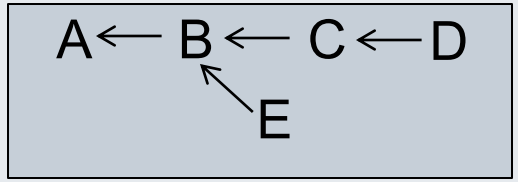
  1) If PRO moves Y in a dispute d (path in the tree) and (Z,Y) $\in \mathcal{A}tt$ then OPP moves Z against Y

  2) If OPP moves Y in a dispute d and (Z,Y) $\in \mathcal{A}tt$ then PRO moves Z against Y only if PRO has not already moved Z *in d*

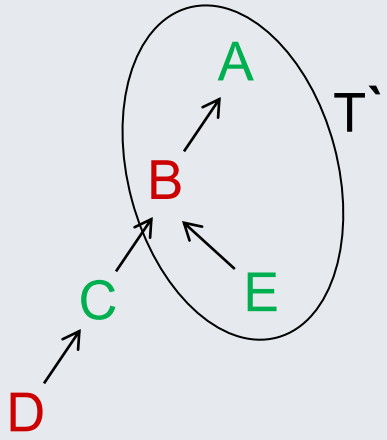
  3) There is a subtree T` of T such that every OPP argument in T' is attacked by a PRO argument (T` is a called a *winning strategy*)

Argument games – winning strategies

AF =



T



PRO wins

Even though not every dispute in T is won by PRO there is a winning strategy T` in which every OPP argument is counter-attacked by PRO

Defining the Preferred Game

- X is in a preferred labelling of < *Args,Attacks* > iff

  There exists a game tree T with root X such that:

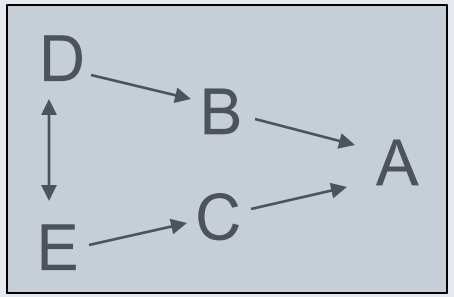
  1) If PRO moves Y in a dispute d (path in the tree) and (Z,Y) $\in \mathcal{A}tt$ then OPP moves Z against Y only if OPP has not already moved Z *in d*

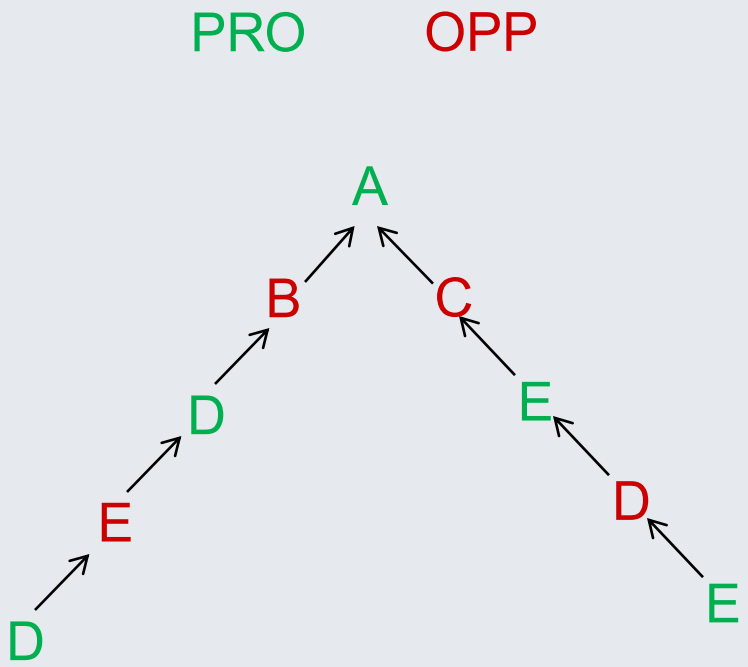
  2) If OPP moves Y in a dispute d and (Z,Y) $\in \mathcal{A}tt$ then PRO moves Z against Y

  3) There is a subtree T` of T such that every OPP argument in T' is attacked by a PRO argument and the arguments moved by PRO in T` do not attack each other (T` is a called a *winning strategy*)

Another Example – Is A in a preferred labelling ?

AF =



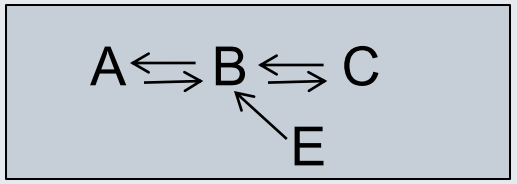


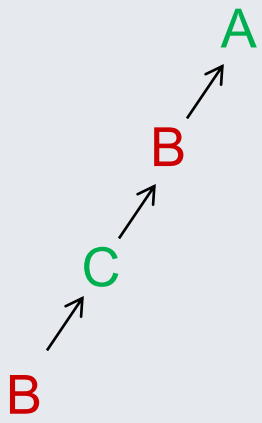
Should PRO win ? -- **NO** according to rule 3 in the def. of the preferred game, the above game tree is not a winning strategy

More Examples

Is A in the grounded labelling ?

AF =



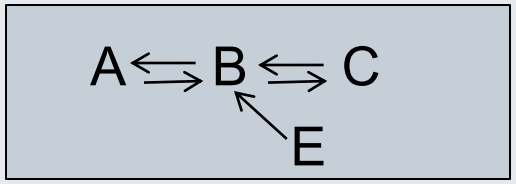


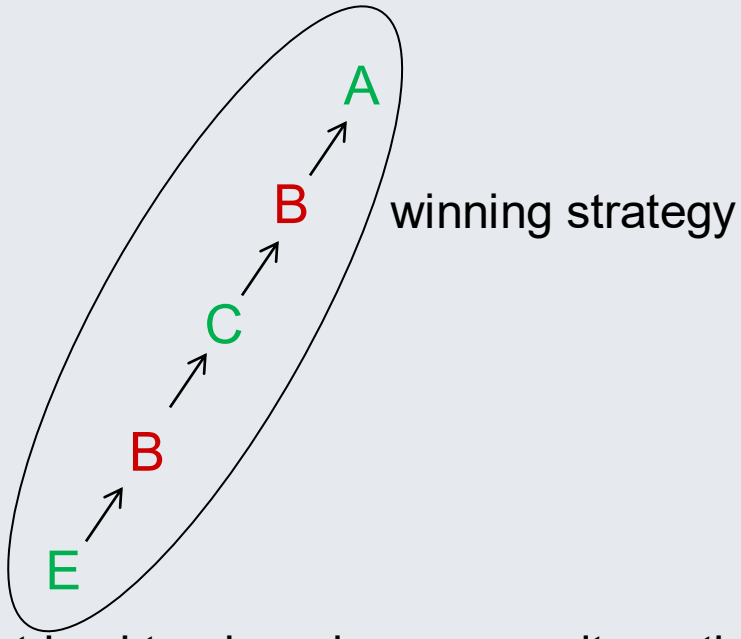
PRO cannot move A or C again

More Examples

Is A in the grounded labelling ?

AF =



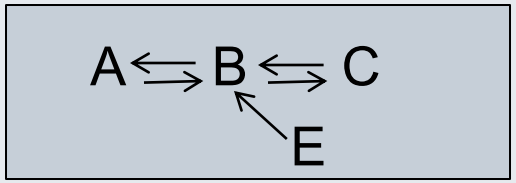


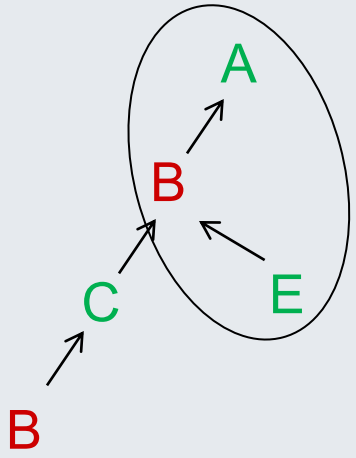
But PRO can move E, and OPP cannot backtrack and move an alternative argument

More Examples

Is A in the grounded labelling ?

AF =





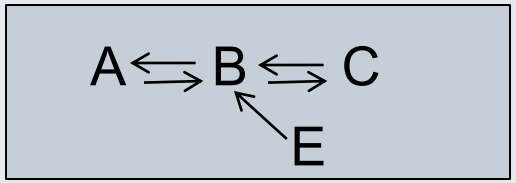
winning strategy

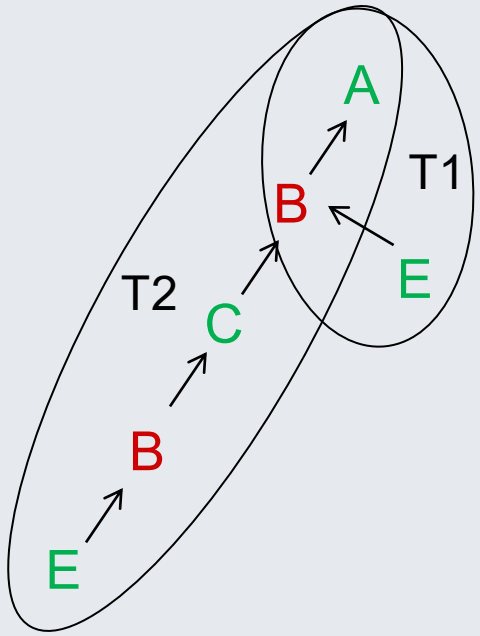
Alternatively, PRO could have backtracked and move E to define a different (more efficient) winning strategy

More Examples

Is A in the grounded labelling ?

AF =





winning strategy T1 =
A ← B ← C

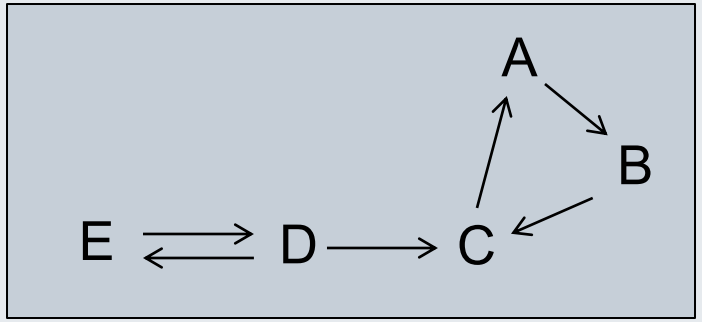
winning strategy T2 =
A ← B ← C ← B ← E

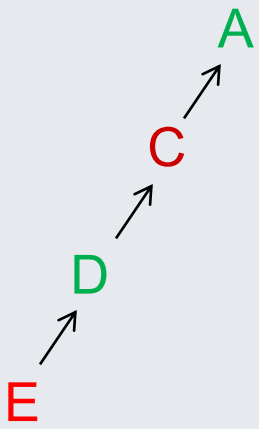
So, when asked to write out game tree and identify winning strategies, write out **whole** game tree and show **all** possible winning strategies (ie., all possible ways that game could have evolved)

More Examples

Is A in the grounded labelling ?

AF =

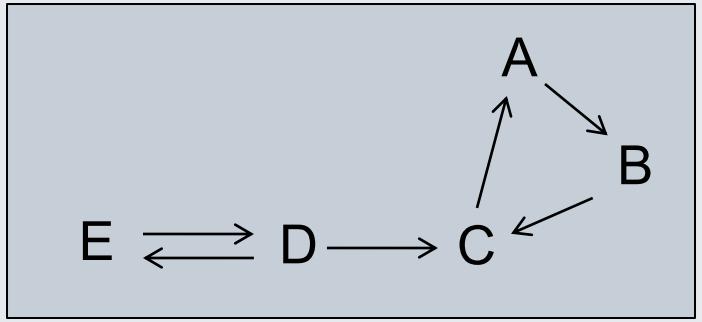


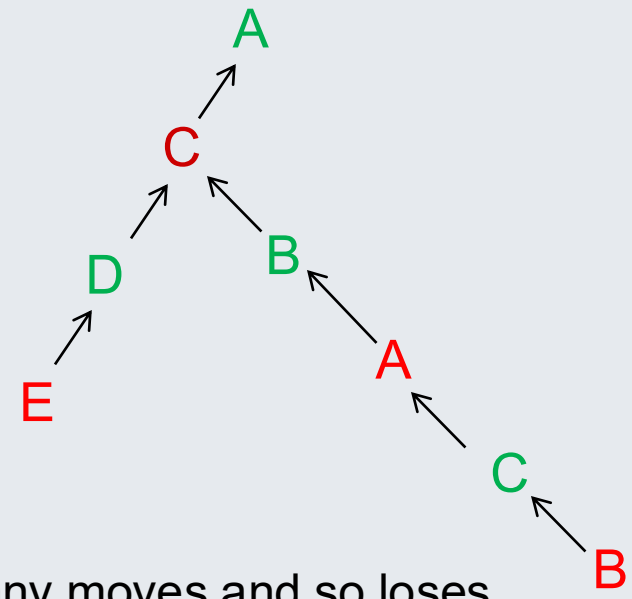


More Examples

Is A in the grounded labelling ?

AF =



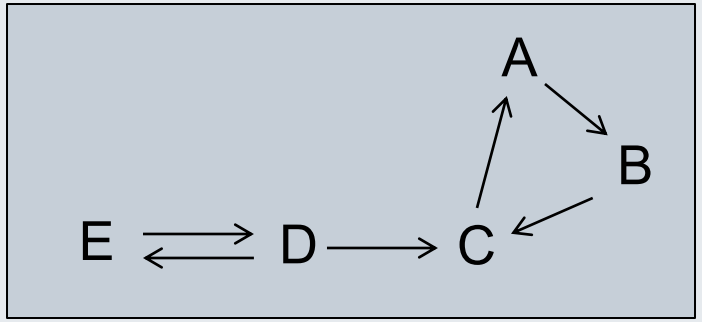


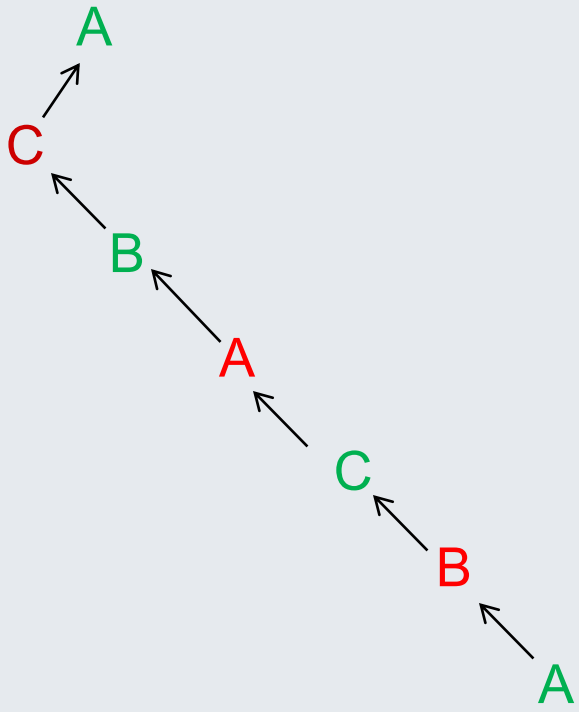
PRO cannot make any moves and so loses

More Examples

Is A in a preferred labelling ?

AF =



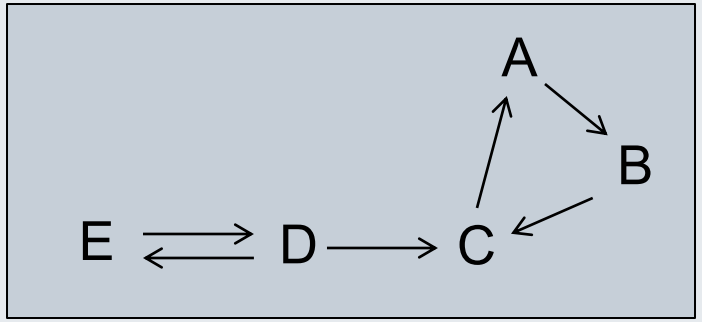


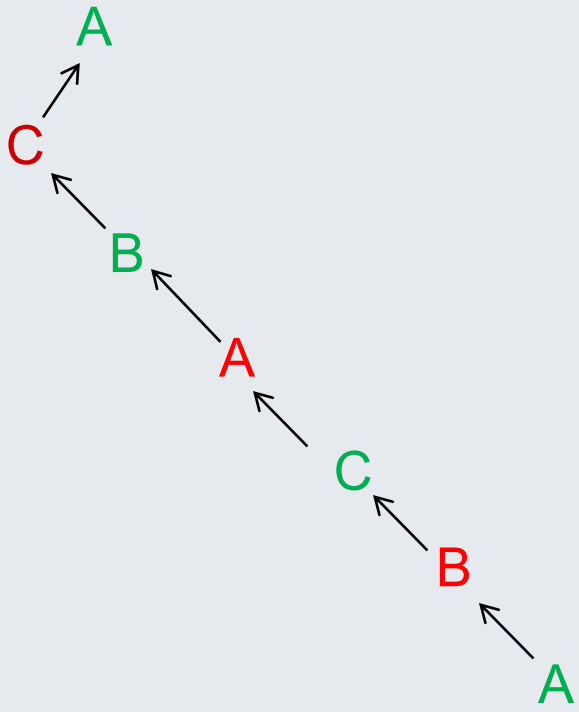
Is this a potential winning strategy for PRO ?

More Examples

Is A in a preferred labelling ?

AF =



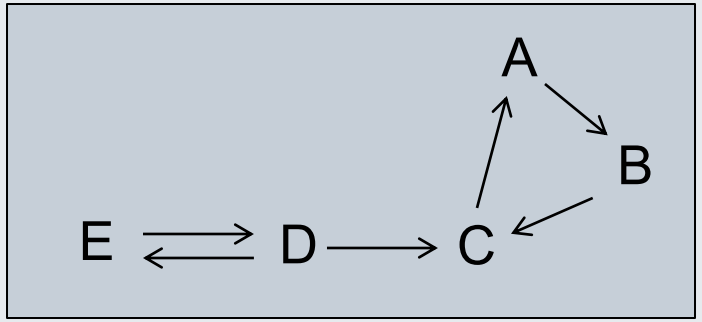


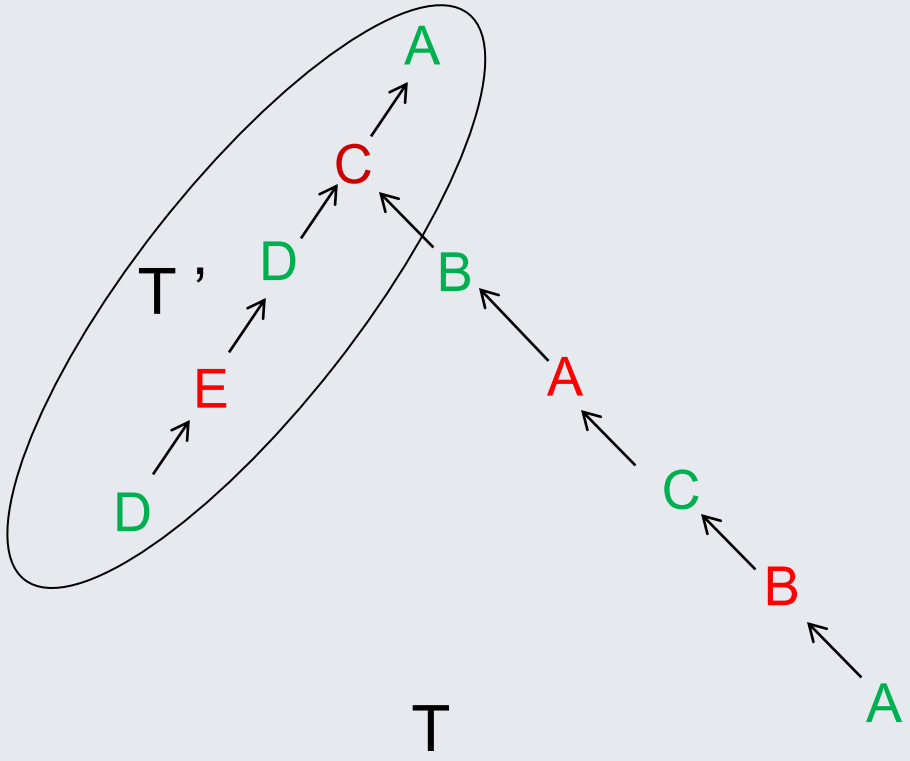
Is this a potential winning strategy for PRO ? No, because PRO has moved arguments that attack each other

More Examples

Is A in a preferred labelling ?

AF =



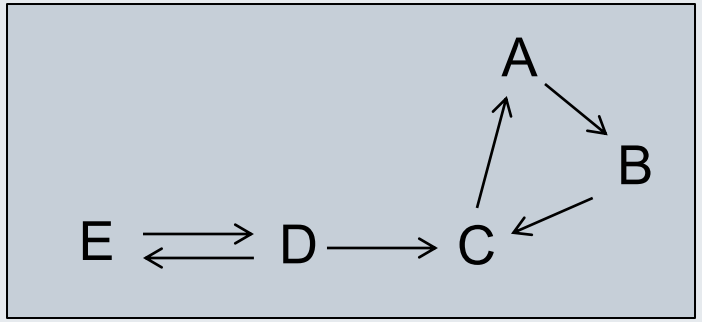


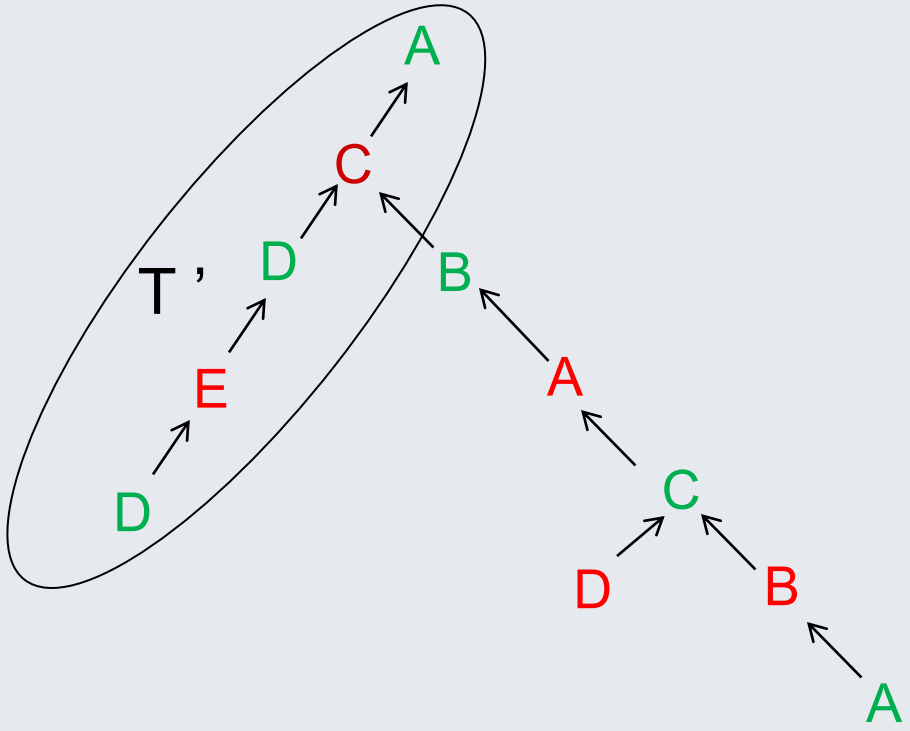
So PRO backtracks to move D against C and establishes a winning strategy T ‘

More Examples

Is A in a preferred labelling ?

AF =



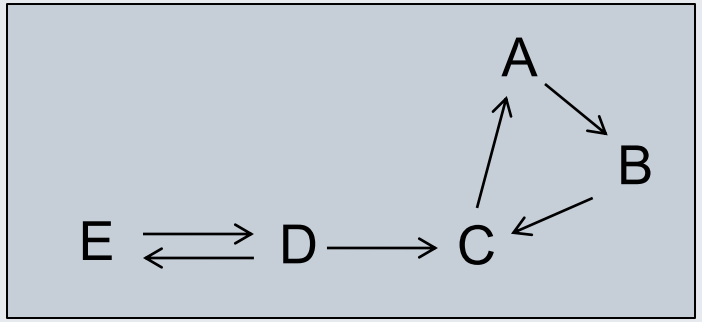


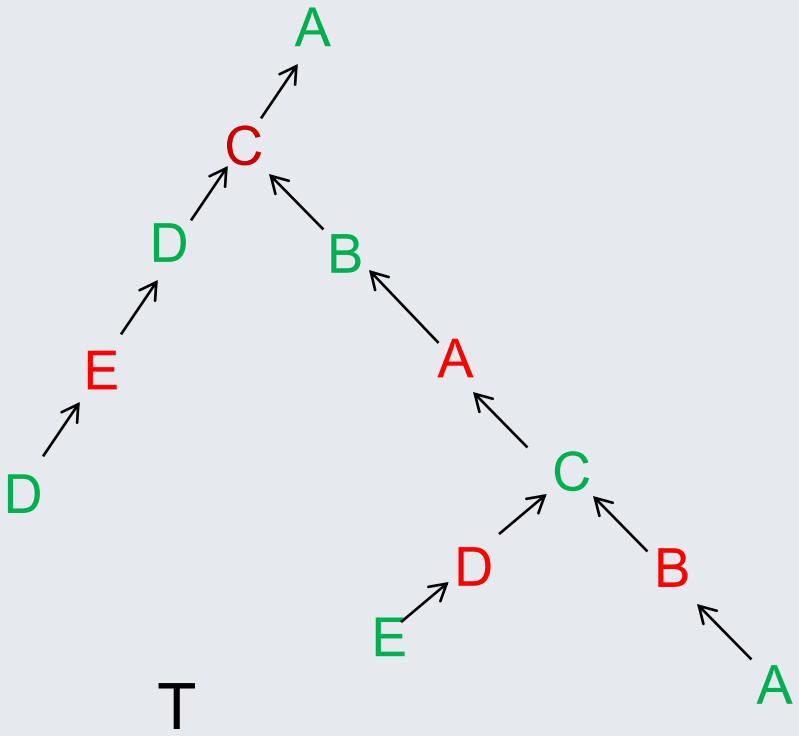
Notice that given the definition of a game, OPP should move D against C,
but this would make no difference to the existence of the winning strategy T ’

More Examples

Is A in a preferred labelling ?

AF =

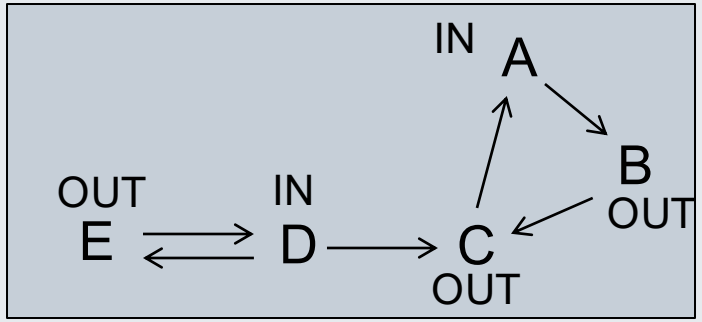


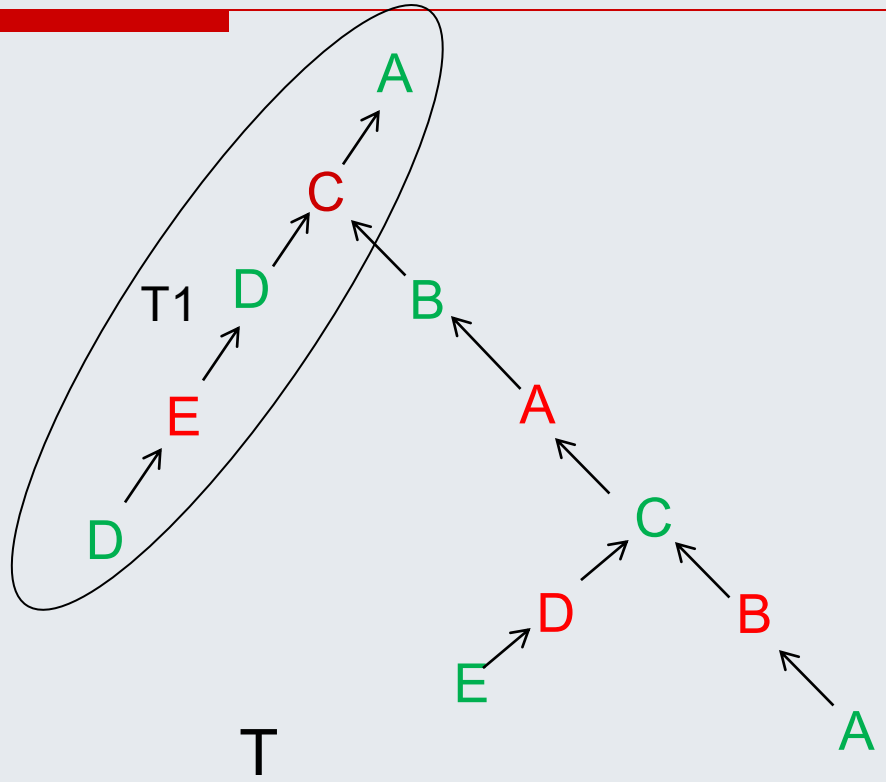


More Examples

Is A in a preferred labelling ?

AF =



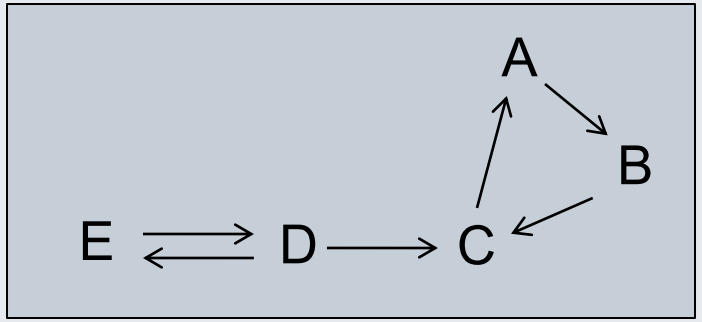


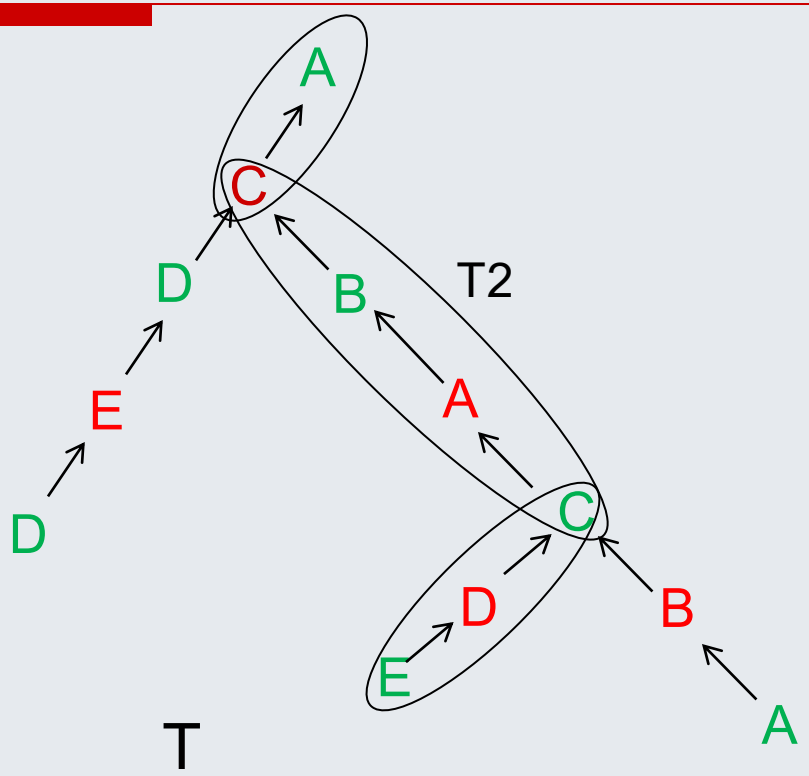
In general write out the whole game tree and then identify all winning strategies (= subtrees in which every OPP argument is attacked by a PRO argument)

More Examples

Is A in a preferred labelling ?

AF =





Is T2 = (A ← C ← B ← A ← C ← D ← E) a winning strategy ?

More on Games

- Note that one can gain efficiency with extra rules on the legality of moves, e.g. prohibiting PRO from moving arguments that attack, or are attacked by arguments PRO has already moved

- Further reading:

- S. Modgil and M.Caminada. Proof theories and algorithms for abstract argumentation frameworks. In : *Argumentation in Artificial Intelligence* (eds. I.Rahwan and G.Simari) 105-129, Springer,2009

(https://cdn.prod.website-files.com/674c48c6202b4f012a14375f/6917492ec9fb21d228577159_ProofTheories_and_Algorithms.pdf)

Week 3 Learning Outcomes

By the end of this week, you will be able to:

- LO1: Identify how to use argument game proof theories to establish whether an argument is in the grounded extension of an argument framework

- LO2: Identify how to use argument game proof theories to establish whether an argument is in a preferred extension of an argument framework

- LO3: Describe how argument game proof theories can be generalized to dialogues for distributed non-monotonic reasoning

- LO4: Describe the importance of these dialogues given recent advances in Artificial Intelligence

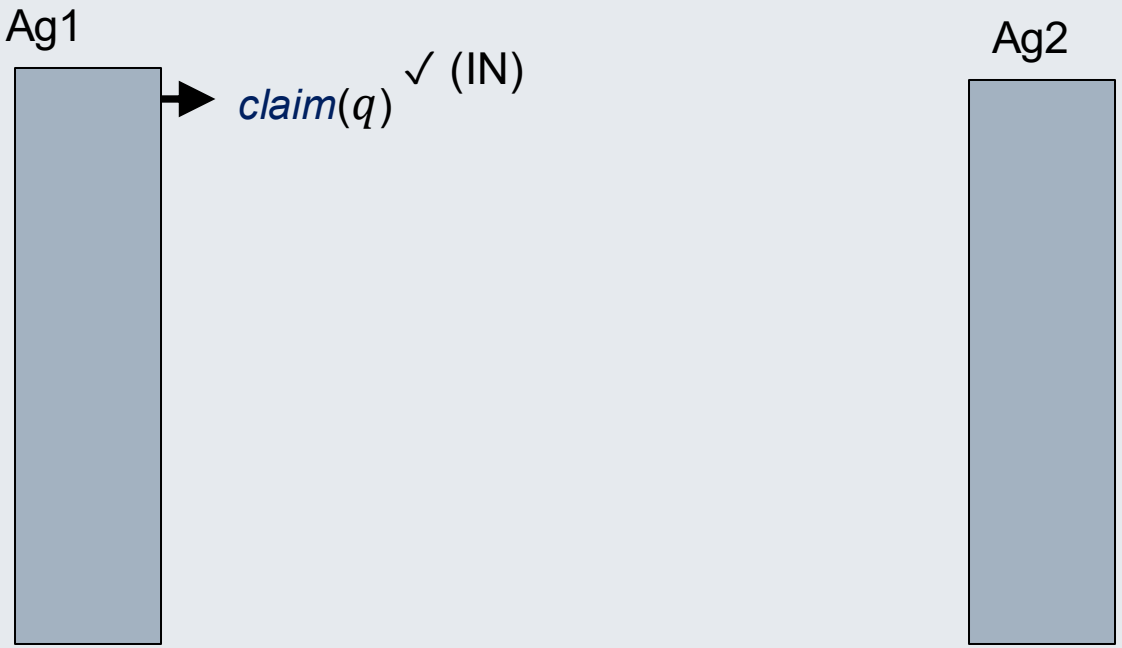
From single agent reasoning to distributed non-monotonic reasoning via dialogue

- When we make utterances in dialogue with each other, these utterances are characterized as locutions

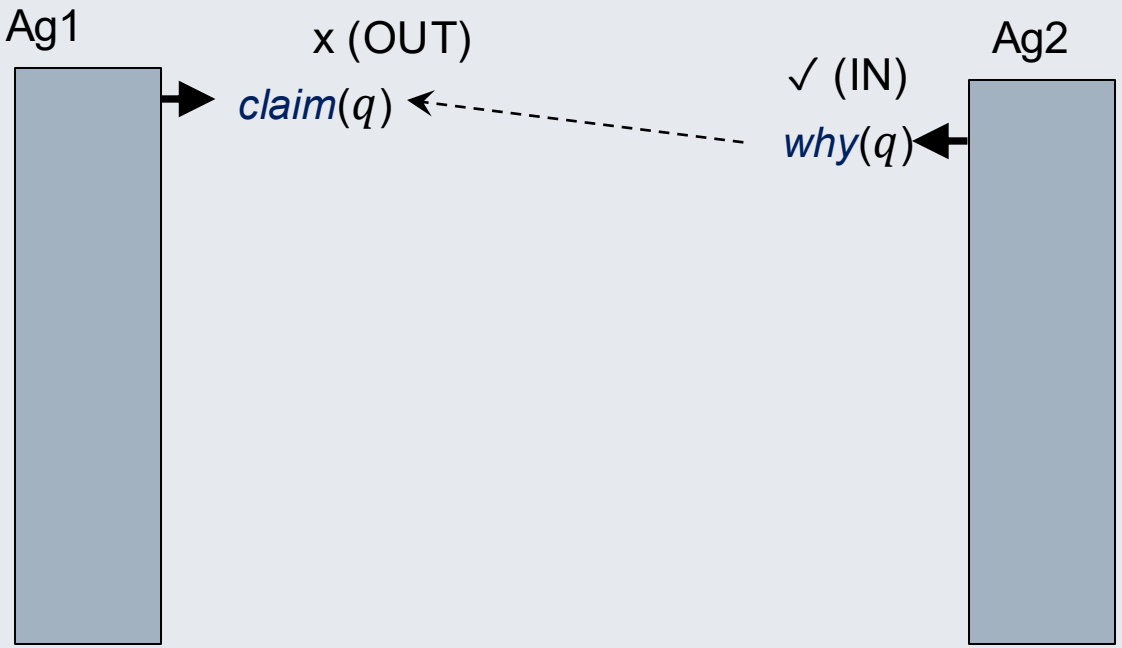
- In dialogues in which we combine our knowledge to collaboratively decide what is the case (what is true) or what to do, we typically make claims, move arguments, challenge each other, reason about preferences or values that help decide amongst decision options

- Dialogue protocols encodes rules on what locutions can be made and what replies can be made. One can then evaluate the status of locutions …

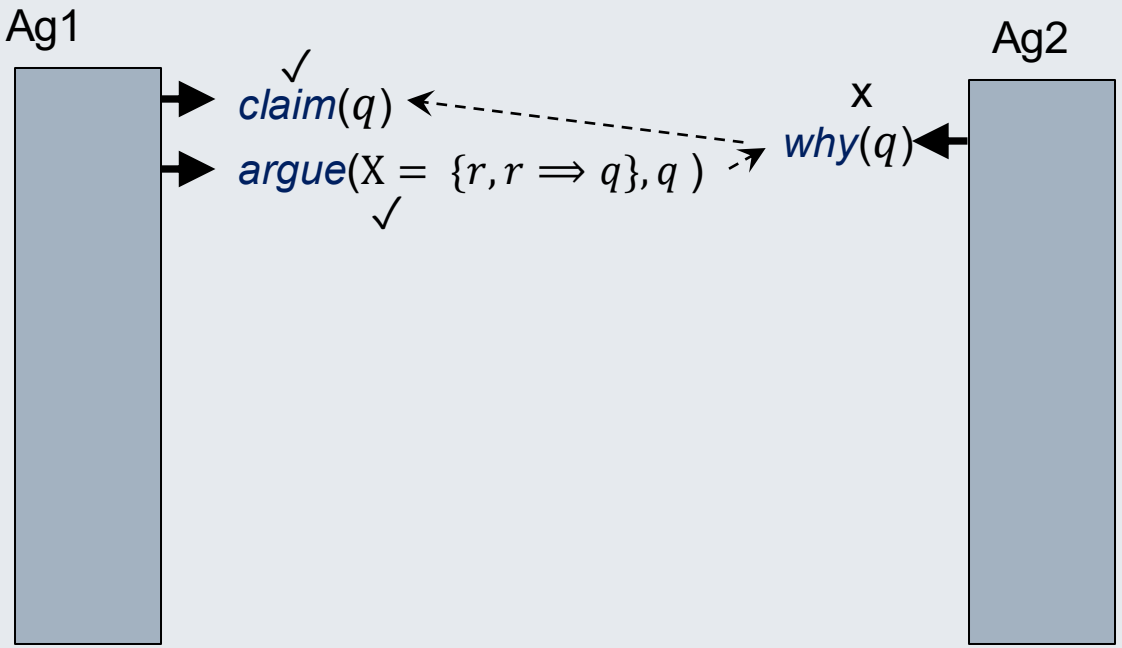
Example



Example



Example



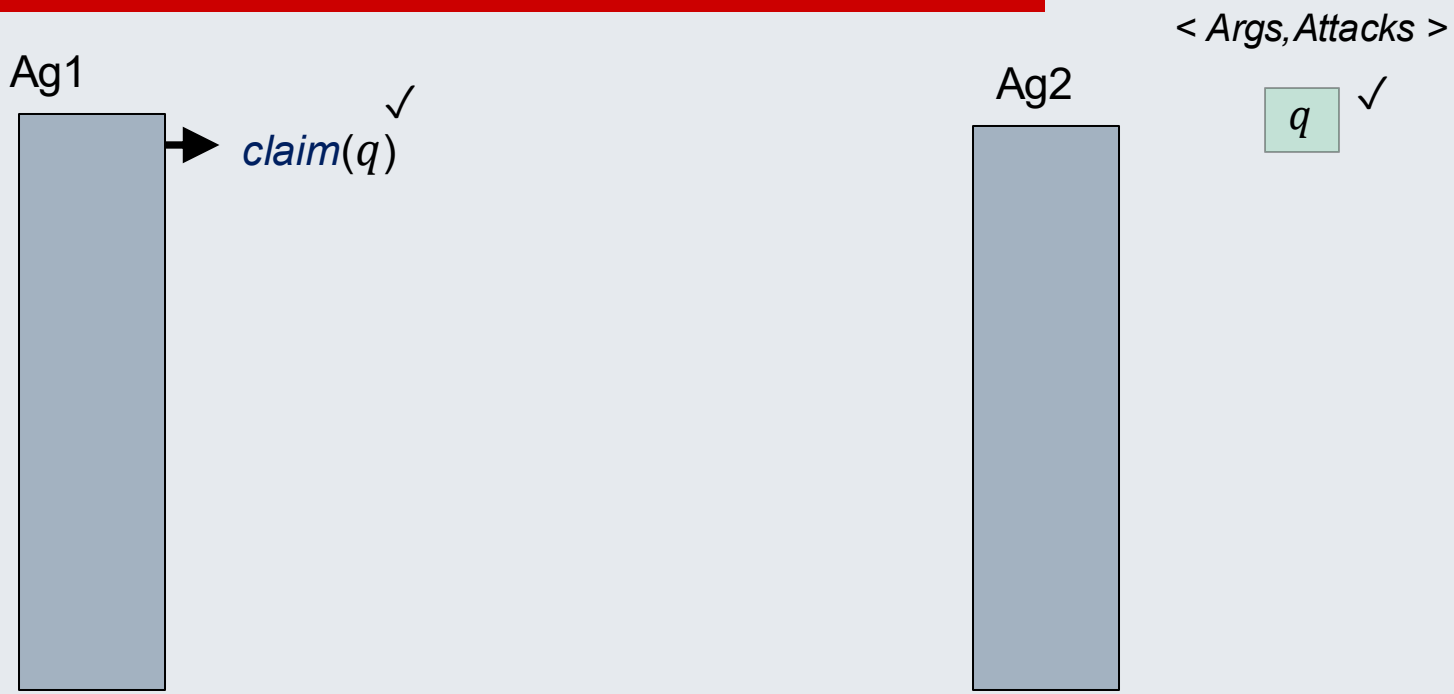
From single agent reasoning to distributed non-monotonic reasoning via inquiry dialogue

- Locutions that make claims, move arguments, challenge, reason about preferences, values, etc
- Dialogue protocol encodes rules on what locutions can be made and what are the legitimate replies that can be made. One can then evaluate the status of locutions
- **Locution Graph** implicitly defines AF = < *Args,Attacks* > and if goal of dialogue is to establish whether *p* is the case (whether *p* is the chosen decision option) we want that:

Soundness and Completeness

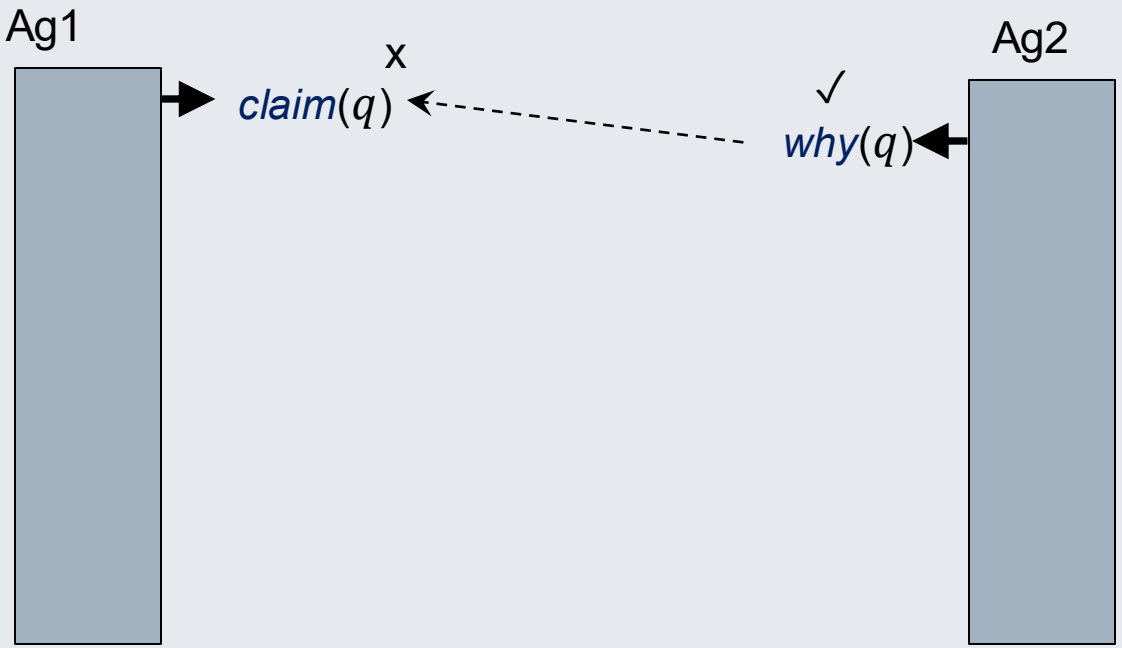
p is winning in locution graph iff *p* is the claim of a justified argument in < *Args,Attacks* >

Example

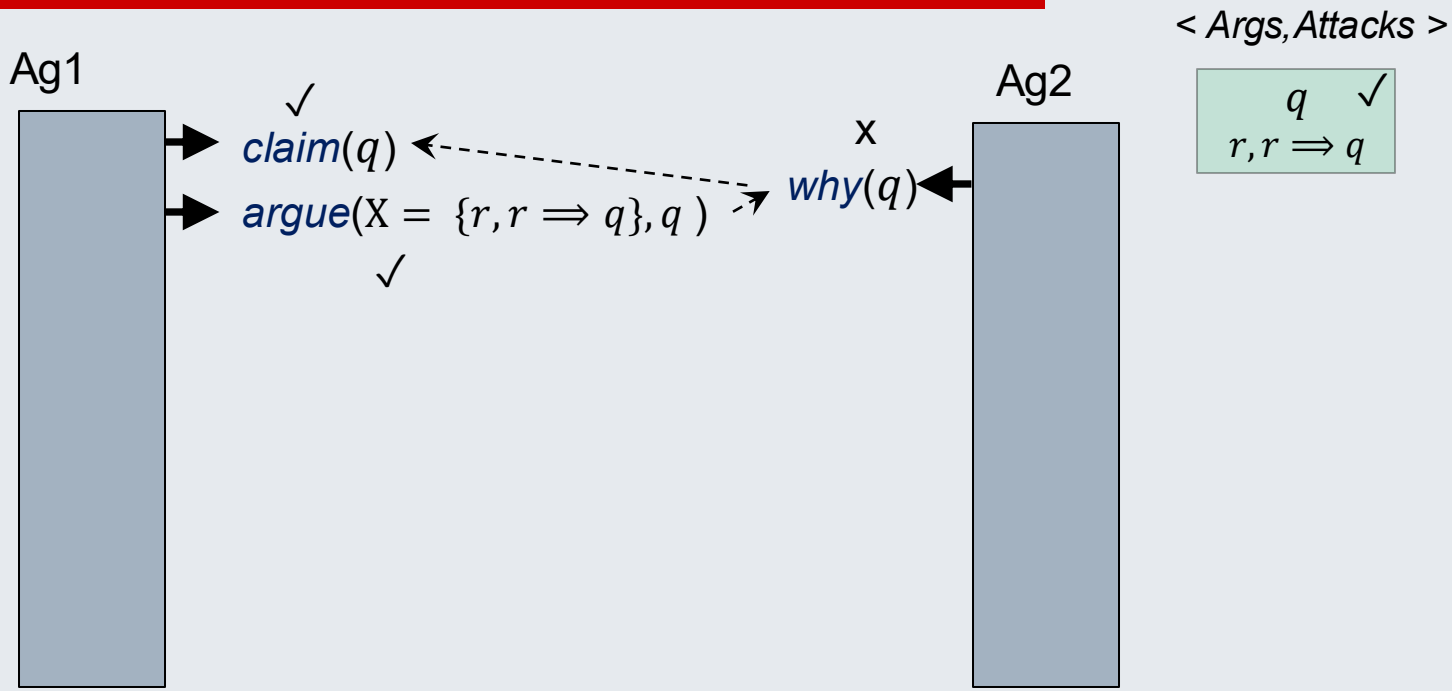


Example

< *Args,Attacks* >



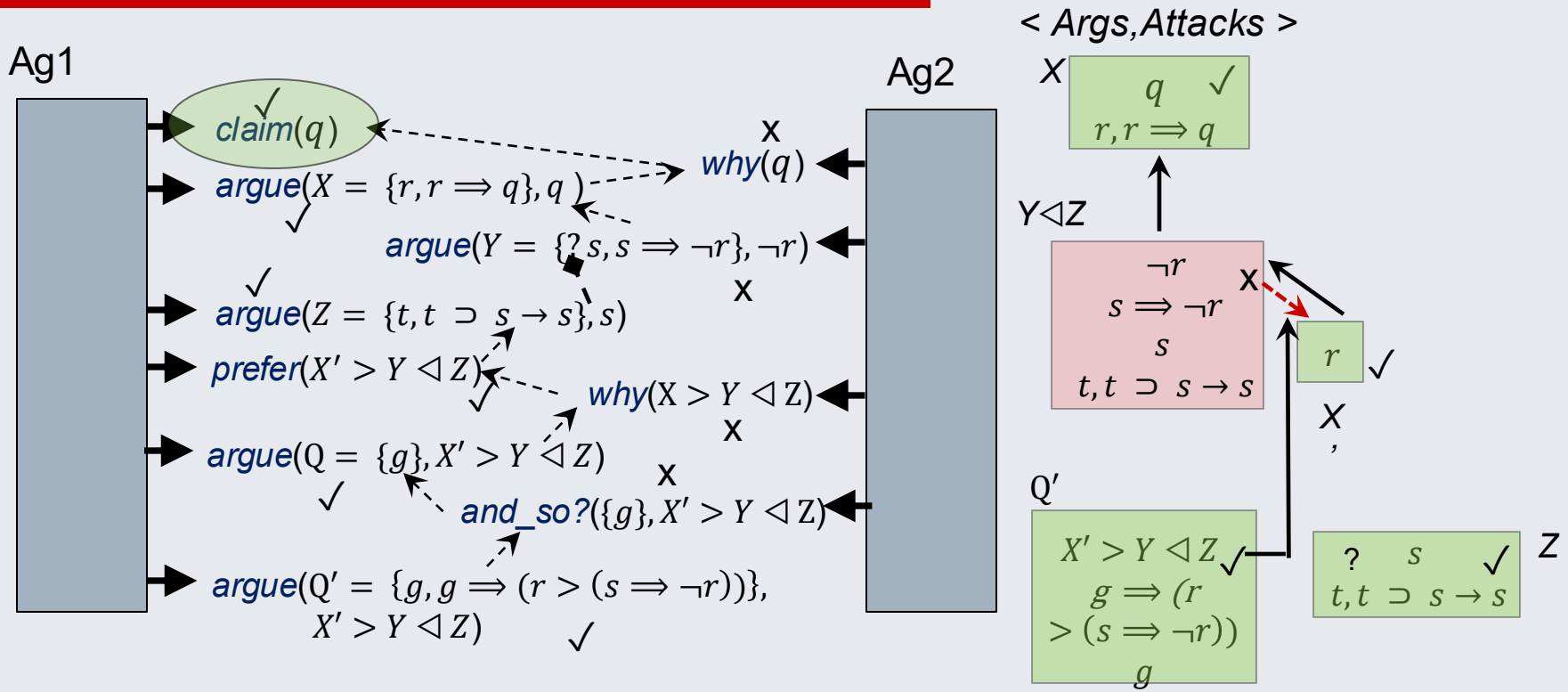
Example



< *Args,Attacks* >

q	✓
$r, r \Longrightarrow q$	

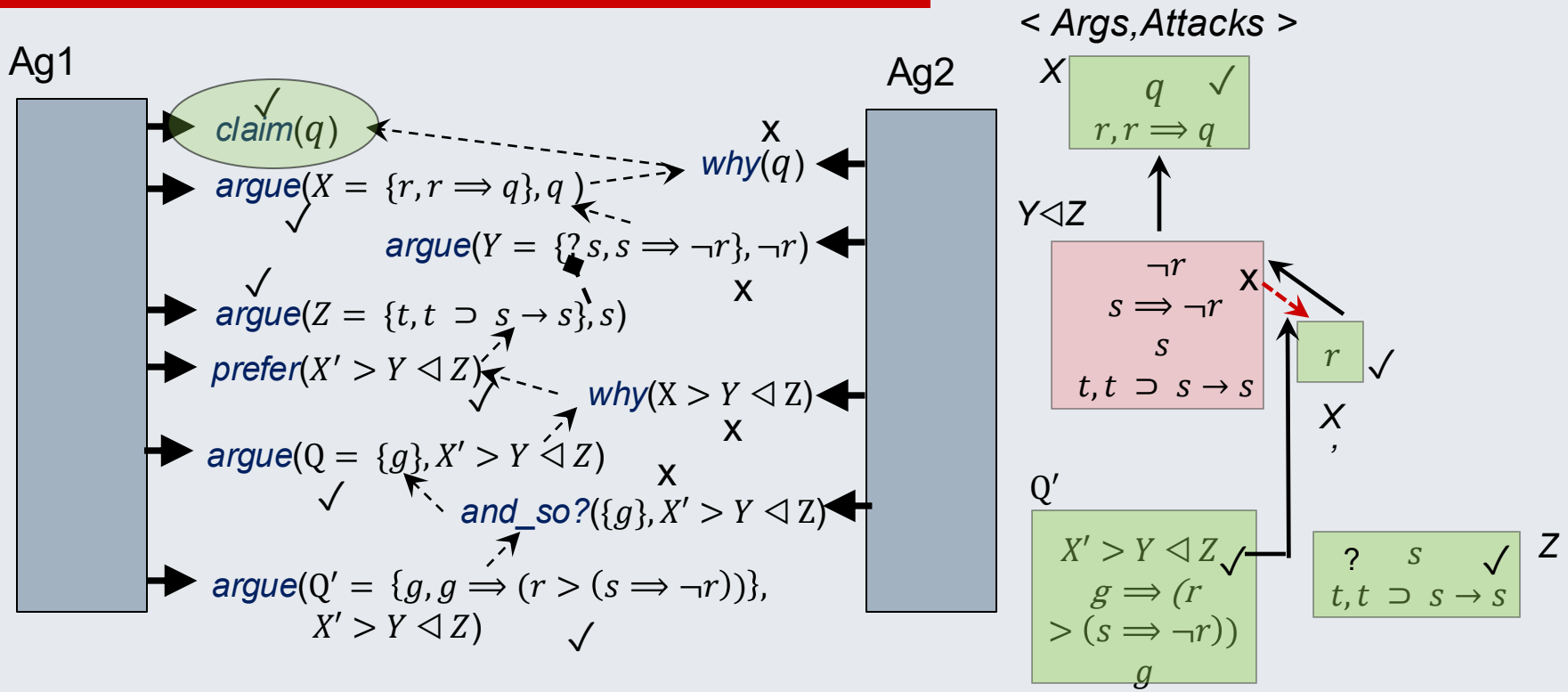
Example



Evaluation of locution graph establishes claim(q) *iff* < *Args,Attacks* > $\mid\!\sim_{Arg}$ q

SC1

Example

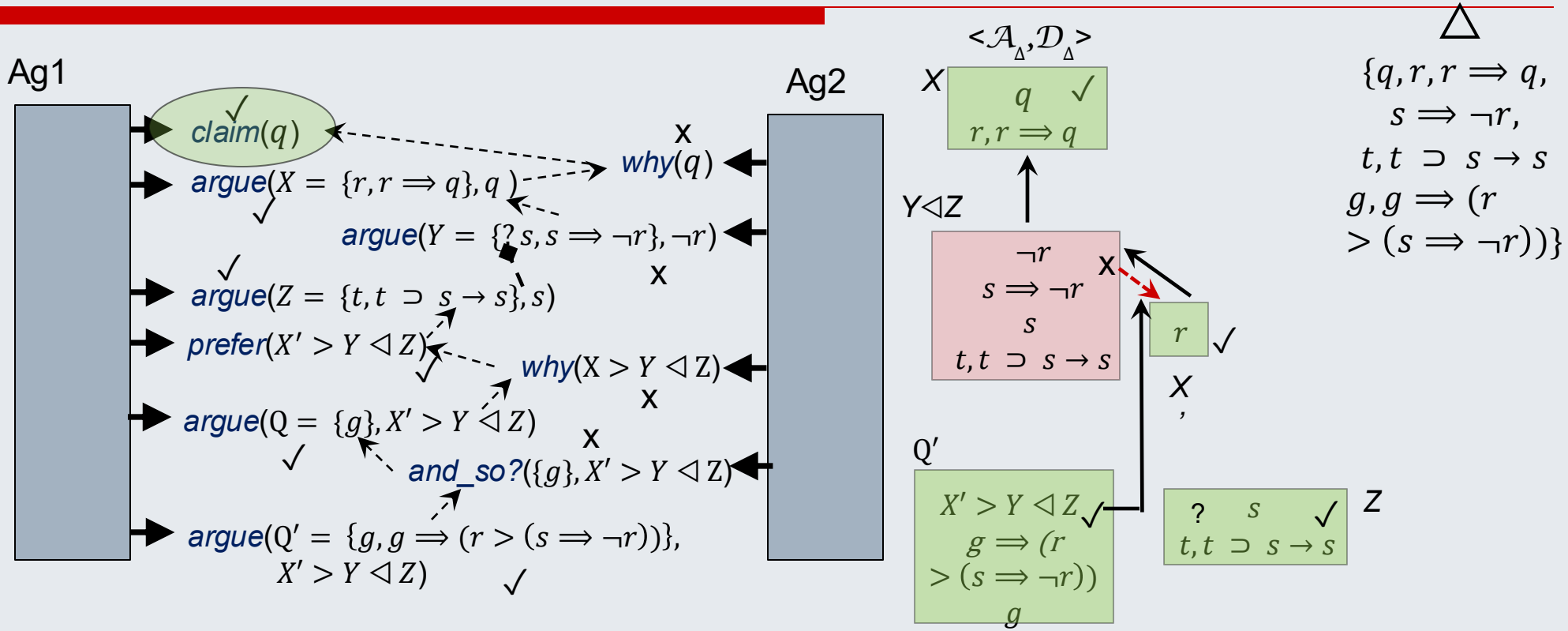


Evaluation of locution graph establishes claim(q) *iff* < *Args,Attacks* > $\mid\!\sim_{Arg} q$ *iff* $\mid\!\sim_{NmL} q$

SC1

SC2

Example



Evaluation of locution graph establishes claim(q) *iff* $< Args, Attacks > \mid\!\sim_{Arg} q$ *iff* $\Delta \mid\!\sim_{NmL} q$

SC

Week 3 Learning Outcomes

By the end of this week, you will be able to:

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- LO2: Identify how to use argument game proof theories to establish whether an argument is in a preferred extension of an argument framework

- LO3: Describe how argument game proof theories can be generalized to dialogues for distributed non-monotonic reasoning

- LO4: Describe the importance of these dialogues given recent advances in Artificial Intelligence

The Value Alignment Problem

- As we delegate more and more tasks to AI systems (LLMs) how can we ensure that their decisions are aligned with (respect) human values ?
- Problems:
  - How can AI systems learn human values and preferences that change over time
  - We often don’t know what our value-based preferences are from the outset. They are often shaped and formed by observation and argument

The Value Alignment Problem

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  - We often don't know what our value-based preferences are from the outset. They are often shaped and formed by observation and argument

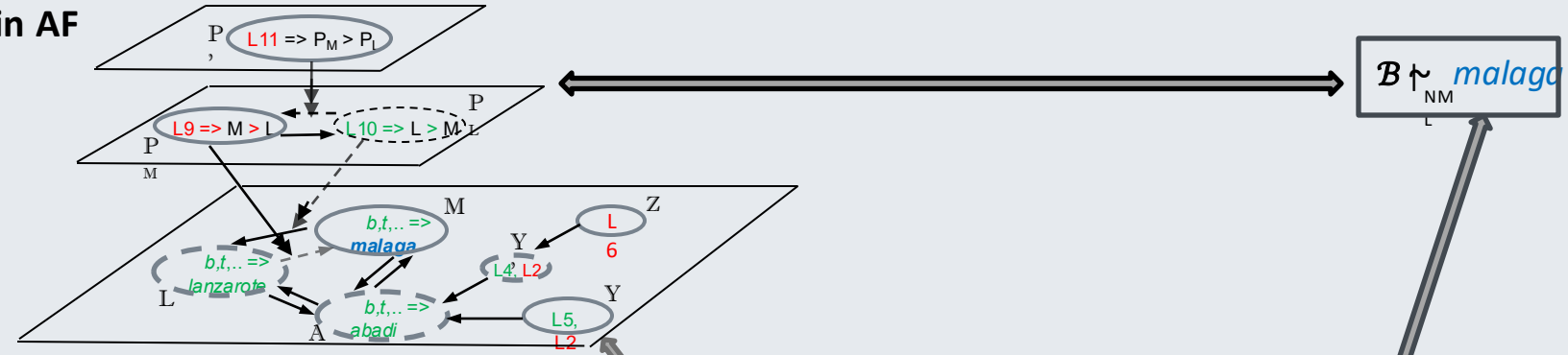
- One solution is to ensure that when AI systems make decisions that have ethical implications, they consult with human users and so:

  1) Learn human preferences
  2) Support humans in helping to shape their preferences

Enculturating AI Agents: Joint Reasoning and Value Alignment Through Dialogue

Malaga **is claim of justified argument in AF**

Malaga **is a nm consequence of contents $\mathcal{B}$ of locutions**



$\mathcal{B} \mid\sim_{NM}$ *malaga*

L1 : Given your budget, (*b*) timing (*t*), landscape (*l*) , weather requirements (r), here are 3 holiday itineraries (𝔸) *Abadi* ….; (𝕄) *Malaga* …; (𝕃) *Lanzarote* …

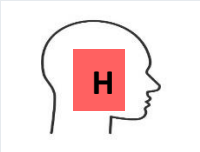
L2 : Reject *Abadi* as unethical labour practices

L3 : Has this been confirmed by experts ?

L6 : The IFA report is fake as evidenced by www ..

L4 : IFA report states Abadi unethical

L5 : Amnesty report states Abadi unethical



PAL
LLM

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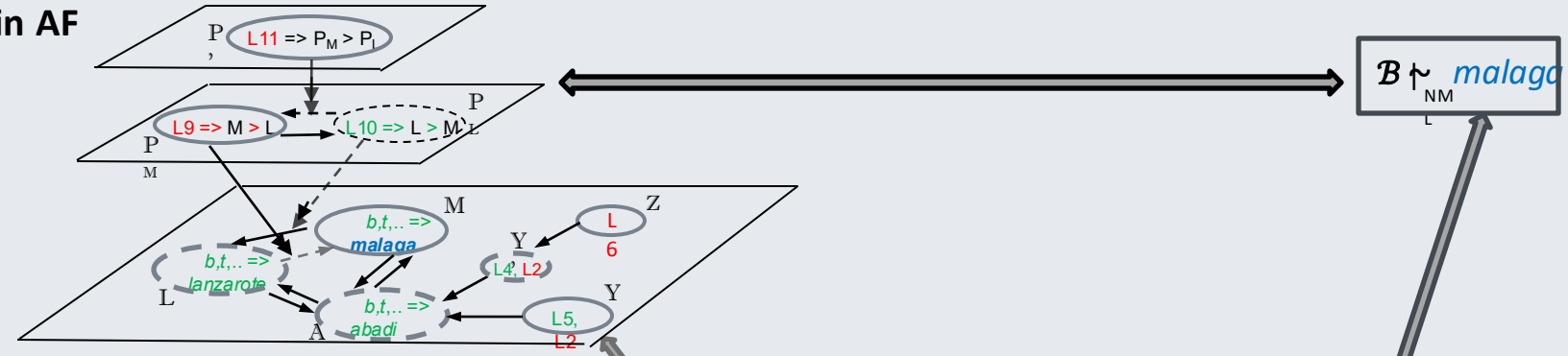

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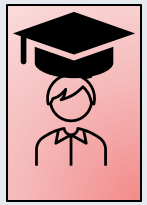
PAL LLM

Shaping human value-based preferences, using arguments jointly constructed with LLM:

- Estimate carbon footprints ?
- Lanzarote greater carbon footprint than Malaga **but** Lanzarote restaurants better and cheaper than Malaga restaurants
- Sustainability more important than restaurant value for money

SocrAPPs for Educational Scaffolding of Critical Thinking Skills

- LLM Socratic apps inculcating critical thinking skills …
- Combatting uncritical information consumption and the epistemic ‘Post truth’ crisis
- Our AI future in which ’menial’ information processing tasks outsourced to AI



Abortion should be illegal

Why ?

Human life is sacred and a foetus is a human

So ?

If a life is sacred it should not be terminated

This implies you are against the death penalty ?

Err, I’m in favour of the death penalty

A human life is sacred unless it takes the life of another

So the executioner’s life is not sacred ?

A human life is sacred unless, unsanctioned by the state, it takes the life of another

Socratic LLM

What you will need to know for exams

- LO1: Identify how to use argument game proof theories to establish whether an argument is in the grounded extension of an argument framework

- LO2: Identify how to use argument game proof theories to establish whether an argument is in a preferred extension of an argument framework

You will not be examined on LO3 and LO4

- LO1: Identify how to use argument game proof theories to establish whether an argument is in the grounded extension of an argument framework

- LO2: Identify how to use argument game proof theories to establish whether an argument is in a preferred extension of an argument framework

~~- LO3: Describe how argument game proof theories can be generalized to dialogues for distributed non-monotonic reasoning~~

~~- LO4: Describe the importance of these dialogues given recent advances in Artificial Intelligence~~